

Assignment no # 02

A cybersecurity analyst is monitoring unusual network traffic patterns. The system represents three types of encrypted packet behaviors as vectors:

$$v_1 = (1,2,3), \quad v_2 = (2,4,6), \quad v_3 = (1,1,0)$$

These vectors model different combinations of packet frequency, size variation, and access timing.

A.

Apply the concept of linear dependence to determine whether the vectors v_1 , v_2 and v_3 are linearly independent or dependent. Justify your answer.

B.

Express the vector v_2 as a linear combination of v_1 and/or v_3 , if possible. Derive the required coefficients.

C.

Use your results to construct a conclusion about redundancy in the packet behavior patterns. Interpret whether all three vectors are necessary for analyzing the system or if some patterns can be eliminated without losing information.

Determine the **span** of the given vectors. Construct a **basis** for this span and calculate its **dimension**.